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A probabilistic metric for comparing metaheuristic optimization algorithms

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ABSTRACT

The evolution of metaheuristic optimization algorithms towards identification of a global minimum is based on random numbers, making each run unique. Comparing the performance of different algorithms hence requires several runs, and some statistical metric of the results. Mean, standard deviation, best and worst values metrics have been used with this purpose. In this paper, a single probabilistic metric is proposed for comparing metaheuristic optimization algorithms. It is based on the idea of population interference, and yields the probability that a given algorithm produces a smaller (global?) minimum than an alternative algorithm, in a single run. Three benchmark example problems and four optimization algorithms are employed to demonstrate that the proposed metric is better than usual statistics such as mean, standard deviation, best and worst values obtained over several runs. The proposed metric actually quantifies how much better a given algorithm is, in comparison to an alternative algorithm. Statements about the superiority of an algorithm can also be made in consideration of the number of algorithm runs and the number of objective function evaluations allowed in each run.

1. Introduction

In recent years, a myriad of evolutionary metaheuristic optimization algorithms have been proposed in the literature. Some algorithms proposed in the nineties can now be called "classics", such as Genetic Algorithms [30], Particle Swarm Optimization [27], Ant Colony Optimization [9] and Harmony Search [14]. In the past decade, many new algorithms were proposed; an incomplete list includes Big Bang-Big Crunch Algorithm [21], Ray Optimization [23], Imperialist Competitive Algorithm [1,22], Mine Blast [36], Firefly [40,12,32], Bat-Inspired [20], Cuckoo Search [13], Dolphin Echolocation [24], Teaching-Learning-Based Optimization [7], Search Group Algorithm [17], Backtracking Search [5,37,38], to name just a few.

Evolutionary meta-heuristic algorithms are popular at searching for the global optimum in non-convex problems [2,15]. A population of particles is initialized over the search domain, and exploits it in a collaborative, interactive manner, looking for the global[1] minimum [25,17]. The initial distribution over the design domain, and the interactions between particles are controlled by random numbers, which provides diversity and robustness to the algorithms. This also makes each run of the algorithms unique. The stream of random numbers used in one run can be controlled by the seed of the random number generator; however, finding the global minimum should not depend on the seed used. Hence, in practice, several runs of the algorithm are required. Ideally, the global minima should be found for every run, but this is usually not the case. The probability (or relative frequency that a particular algorithm finds the global minimum can be measured by the number of times this happens, relative to the total number of runs. This measure is not straightforward because in many runs the algorithm converges to local or to "near global" minima. In this setting, comparing the performance of different algorithms becomes a relevant problem.

When a new metaheuristic algorithm is proposed, benchmark problems should be used to compare its performance to existing algorithms. The same is true for application of an existing algorithm to a new field. In the early days, such comparisons were severely handicapped. Taking the example of truss structures optimization, early works typically presented only the best design found in several runs [35,19,6,39]. Strikingly, sometimes not even the number of objective function evaluations was reported. Such limited, biased comparisons are not acceptable these days [18]. More recently, the performance of metaheuristic algorithms has been compared by

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[1] It is well known that solving non-convex problems using heuristic algorithms yields no warranty that the resulting minimum is indeed a global minimum. This issue is beyond the scope of this article and is not further addressed herein.

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evaluating statistics of the minima obtained in each of several runs of an algorithm. Statistics include the mean value, the standard deviation, the best (smallest) and the worst largest) minima obtained in several runs. Examples in truss optimization include [31], Fadel Miguel et al. [11], Hasançebi et al. [20], Gandomi et al. [13], Kaveh & Farhoudi [24], Degertekin & Hayalioglu [7], Kaveh et al. [26], Gonçalves et al. [17] and Carraro et al., [4].

A problem with the above metrics is that they are not unique. Surely, for minimization problems, the mean, the standard deviation, the best and the worst minima found in several runs should all be as small as possible. Ideally, the standard deviation should be zero and the mean should be the smallest possible, in which case the other metrics (best and worst) would agree with the global minima of the problem. However, if the mean or best minima found using algorithm A is marginally smaller than those found using algorithm B, but the standard deviation is marginally larger, which algorithm is better? The cited metrics alone cannot answer this question. Two compounding problems related to the above are that: a) the convergence of the above statistics with the number of runs is usually not analyzed; and b) the number of runs varies significantly from one paper to another.

More robust comparison procedures have been proposed and are employed in other fields, *e.g.* Kolmogorov-Smirnov, Shapiro-Wilk and D'Agostino-Pearson, Wilcoxon Signed-Rank Tests [8,5,10,33]. These have the disadvantage of being more complex, and assuming normality of the analyzed data.

In this context, the main contribution of this paper is the proposal of a novel metric for comparing metaheuristic optimization algorithms. It is based not only on averages, standard deviations and best/worst results, but on the entire information concerning minimum objective function values obtained during executions of the algorithms. Based on the idea of statistical interference, it computes which method, among two being compared, has larger probability of obtaining the best result if just one execution was performed. The proposed metric is very simple to compute, and it is non-parametric, in the sense that it does not require any assumption about probability distributions.

The remainder of this paper is organized as follows. In Section 2, the proposed probabilistic metric is presented and explained. Section 3 presents numerical results for three benchmark example problems. Results include convergence plots of the proposed metric, and of usual metrics, w.r.t. number of algorithm runs. Some conclusions are drawn in Section 4.

2. Proposed probabilistic metric

Consider a general optimization problem, defined as:

$$\begin{aligned}
&\text{Find } \mathbf{d}^* \\
&\text{which minimizes } O(\mathbf{d}) \\
&\text{subject to :} \\
&\quad h_i(\mathbf{d}) = 0, i = 1, \ldots, p; \\
&\quad g_j(\mathbf{d}) \leqslant 0, j = 1, \ldots, q; \\
&\quad \mathbf{d} \in S \subset \mathbb{R}^n,
\end{aligned} \tag{1}$$

where $\mathbf{d}^* \in \mathbb{R}^n$ is a vector of design parameters, which minimizes an objective function $O(d)$, subject to p equality and q inequality constraints, and $S = \{\mathbf{d}_{\min}, \mathbf{d}_{\max}\}$ is a set of side constraints.

Assume an author is proposing a new heuristic algorithm[2] to solve the optimization problem in Eq. (1). To demonstrate effectiveness of his algorithm, he will have to compare his solution to that of existing algorithms. Since, in general, heuristic algorithms depend on random numbers, the above comparison needs to be done for several runs $k = 1, \ldots, m$ of each algorithm. Let $y_P^k = O(\mathbf{d}^*)$ denote the minimal value of the objective function, found by the ***P***roposed algorithm, in the k-th run, and let $y_E^k = O(\mathbf{d}^*)$ denote the same metric for an ***E***xisting algorithm. Assume also the random parameters governing each algorithm are independently generated from run to run. After several runs of each algorithm, optimal objective functions values are collected in vectors $\mathbf{y}_P = [y_P^1, y_P^2, \ldots, y_P^{m_P}]$ and $\mathbf{y}_E = [y_E^1, y_E^2, \ldots, y_E^{m_E}]$. Now, each component of vectors $\mathbf{y}_P$ or $\mathbf{y}_E$ can be seen as realizations of identically distributed random variables Y_P and Y_E, whose empirical probabilities of occurrence are given by $1/m_P$ and $1/m_E$, respectively.

If the probability density function $f_{Y_E}(y_E)$ of the minimum values Y_E obtained by the existing algorithm and the cumulative distribution function $F_{Y_P}(y_P)$ of Y_P were known, then the probability that the proposed algorithm produces an objective function value Y_P smaller than Y_E would be given by:

$$P_{better} = P[\{Y_P < Y_E\}] = \int_{-\infty}^{+\infty} f_{Y_E}(y) F_{Y_P}(y) dy \tag{2}$$

where P_{better} can be read as "the probability that, in a single run, the proposed algorithm yields a smaller (global?) minimum than the existing algorithm". Interpretation of this probability is straightforward. For instance, if the proposed algorithm has a 50% probability of producing better results than the existing method, their performances are equivalent. If this probability is larger than 50%, then the proposed algorithm outperforms the existing algorithm. The proposed probability metric also indicates how much better the performance of one algorithm is in comparison to another. A probability of 99%, for instance, gives much more confidence in the performance of the proposed algorithm, relative to the existing algorithm, than a probability just above 50%.

In general, the probability distribution functions in Eq. (2) are not known. Nevertheless, non-parametric empirical distributions, derived exclusively from observed vectors $\mathbf{y}_P$ and $\mathbf{y}_E$, can be employed to compute the proposed probability metric. The empirical approximations to the required probability density and cumulative distribution functions are given, respectively, by:

$$f_{Y_E}(y_E^k) \cong \frac{1}{m_E} \tag{3}$$

$$F_{Y_P}(y_E^k) \cong \frac{1}{m_P} \sum_{j=1}^{mP} I\left(y_P^j \leqslant y_E^k\right) \tag{4}$$

where $I()$ is the indicator function, resulting one (1) when the operand is true, zero otherwise. The integral presented in Eq. (2) can now be estimated, in a Monte Carlo sense, by:

$$P_{better} = P[\{Y_P < Y_E\}] \cong \frac{1}{m_P m_E} \sum_{k=1}^{m_E} \left(\sum_{j=1}^{mP} I(y_P^j < y_E^k) \right) \tag{5}$$

which asymptotically approaches Eq. (2) as $m_P \to \infty$ and $m_E \to \infty$.

For problems involving discrete design variables, many of the optimal solutions will be the same. It would be unfair to claim the proposed algorithm to be better, if it is producing the same results. Hence, for these problems it is convenient to also evaluate the probability that both algorithms are equivalent:

$$P_{eq} = P[\{Y_P = Y_E\}] \cong \frac{1}{m_P m_E} \sum_{k=1}^{m_E} \left(\sum_{j=1}^{mP} I(y_P^j = y_E^k) \right) \tag{6}$$

Eq. (6) can also be used when a numerical tolerance is considered for the inequalities in Eq. (5). The probability that the *P*roposed algorithm is worse than the *E*xisting algorithm is:

[2] The above discussion illustrates use of the proposed metric by an author trying to defend a newly proposed algorithm, which is a usual application. The proposed metric, however, can also be used for the uninterested comparison between any two existing algorithms *A* and *B*.

$$P_{worse} = P[\{Y_P > Y_E\}] \cong 1 - P_{better} - P_{eq} \quad (7)$$

In practice, m_P and m_E in Eqs. (5) and (6) need to be "large". Discussion about sample size representativeness has been the object of many papers in the literature (*e.g.*, [28,3,29]. In a simplified way, since the probability approximated by Eq. (5) is not strongly dependent on the tails of the distributions involved, the required number of runs can be determined by estimating the standard error of the mean of vectors Y_P and Y_E, as results of new runs are included. The sample can be considered large enough if the estimated standard error is smaller than a given tolerance. Alternatively, convergence plots of the probability P_{better} w.r.t. the number of runs can indicate when the computed probability no longer depends on the particular stream of random numbers. This also indicates sufficiency of m_P and m_E, and is the approach adopted herein.

One of the advantages of the probabilistic metric proposed above is that the numbers of runs can be different ($m_P \neq m_E$), which is useful when comparing to an algorithm run by another author, although the difference should not be too large.

One very significant advantage is that the metric in Eq. (5) is non-parametric, i.e., it does not assume any probability distributions to the random variables Y_P and Y_E. This advantage should not be underestimated, as in complex non-convex problems, heuristic algorithms are likely to get stuck in local minima, leading to multi-modal distributions for Y_P and Y_E. The metric proposed in Eq. (5) is computed from the actual observations of the algorithm runs. Herein, we refer to probability distributions only to explain how the proposed metric (Eq. (5)) is derived from the "classical" result in Eq. (2). Eq. (5) can also be interpreted as the classical interference problem, when computed by Monte Carlo Simulation. The probability distributions of Y_P and Y_E are actually not necessary, because Monte Carlo sampling is performed by running the optimization algorithms.

Clearly, the proposed metric can also be used to compare several algorithms. In this case, one of the algorithms is taken as the "reference" algorithm (for instance, the "proposed" algorithm), and the probabilities are computer w.r.t. the reference algorithm.

Finally, the probabilistic metric was presented herein with the point of view of an author trying to showcase a newly proposed algorithm against existing algorithms, which occurs very often in the literature. However, the proposed metric can be used for the uninterested comparison between any two existing algorithms *A* and *B*, in which case the sub-indexes *P* and *E* above are simply replaced by *A* and *B*.

3. Numerical examples

To illustrate application of the proposed P_{better} metric, and compare it to other metrics usually adopted in the literature, three benchmark optimization problems are addressed in this section. The performance of four optimization algorithms is compared: Firefly Algorithm FA, [40], Imperialist Competitive Algorithm ICA, [1,22], Backtracking Search Algorithm BSA, [5] and Search Group Algorithm [17]. In all problems, the SGA algorithm is taken as reference (***P*** algorithm), but we state clearly: we have no preference for any among the cited algorithms. Herein we perform an uninterested comparison between them, where the only interest is in highlighting the advantages of the proposed probabilistic metric. For those problems were the SGA algorithm is indeed better, we present the probability P_{better} directly. For those problems where another algorithm is clearly superior, we present the complement probability (P_{worse}), as this is easier to assimilate.The number of objective function evaluations (OFE) is limited, considering the complexity of each problem, and is the same for all algorithms. Parameters for all algorithms are taken from the literature, where available. Following notation in Miguel et al. [11], the FA algorithm was run using $h = 1 \times 10^8$, $\beta_0 = 1$, $\gamma = 1$, $\alpha = 0.5$ and $\boldsymbol{\varepsilon}_i \sim U[0, 1]$. Following Gonçalves et al. [17], the SGA parameters were taken

Table 1
Design parameters for the eleven-bar truss benchmark example.

Design parameter	Non-SI units	SI units
Modulus of elasticity	10^4 ksi	68.95 GPa
Weight density	0.1 lb/in^3	2767.99 kg/m^3
Allowable stress in tension (σ^t)	25 ksi	172.37 MPa
Allowable stress in compression (σ^c_j)	25 ksi	172.37 MPa
Allowable *y*-displacement	2 in	0.0508 m

Table 2
Parameters of the FA, ICA, BSA and SGA algorithms for Problem 1.

Algorithms	Parameters
FA	n_{pop} = 10, it_{max} = 3000
ICA	$N_{Col} = 367$, $N_{imp} = 2$, $\beta = 1.95$ and $\gamma = \frac{\pi}{4}$
BSA	n_{pop} = 10, it_{max} = 3000, $m_r = 1.0$, $\alpha \sim N(0, 3)$[3]
SGA	$n_{pop} = 430$, $n_{mut} = 5$, $n_g = 19$, $\alpha^0 = 0.67$, $it^{max}_{global} = 0.50$

[3] $N(0, 3)$: normal random value with mean and standard deviation equal to 0 and 3, respectively.

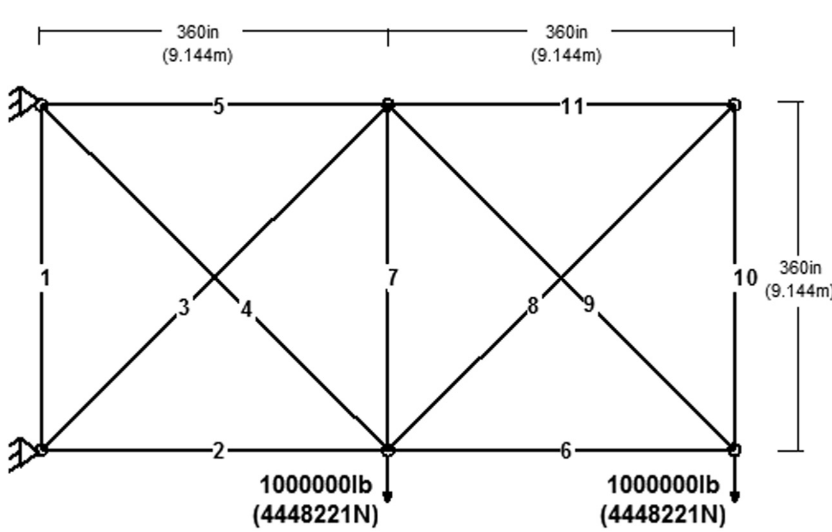

Fig. 1. Eleven-bar truss benchmark example (Problems 1 and 2).

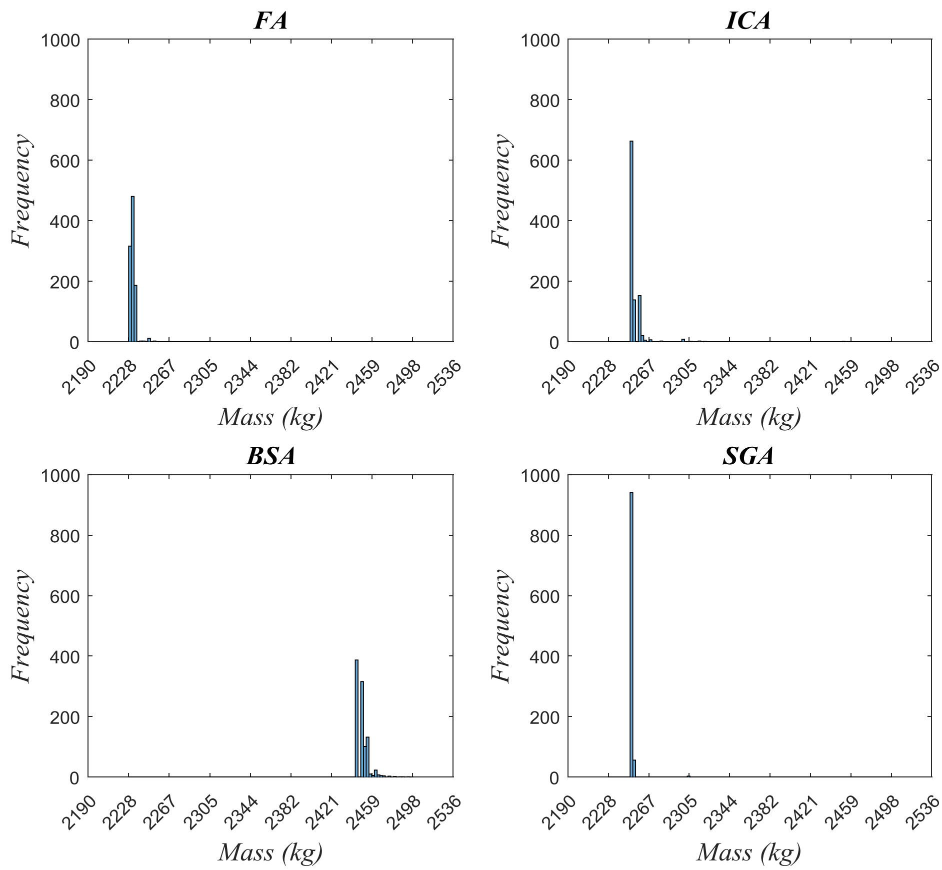

Fig. 2. Histograms of objective function minima computed in 1000 runs of Problem 1.

as: t = 1, 2 and 3 (we employed one of these values of t for the 3 mutated individuals), $h = 5 \times 10^9$, $\alpha_{\min} = 0.08$, $b = \max(1 - 4k/it_{global}^{max}, 0.25 - k/it_{global}^{max})$ and $\boldsymbol{\varepsilon}_i \sim U[-0.5, 0.5]$. Parameters of the SGA and FA that varied from one example to the other as well as the parameters of BSA and ICA are detailed within each example. Finally, for some runs of the FA algorithm, no viable solution was found. For these runs, a large number (10^6) was used to compute the performance statistics.

3.1. Eleven-bar truss

This truss is often used as a benchmark example, and is used herein in two problem variants: Problem 1, size and topology optimization; Problem 2, simultaneous size, shape and topology optimization. The ground structure is shown in Fig. 1 and the design parameters are given in Table 1.

Table 3
Statistics of objective function minima found over 1000 runs of each algorithm.

Algorithm	Best (kg)	Worst (kg)	Mean (kg)	COV (%)
FA	2228.43	2252.68	2231.93	0.14
ICA	2250.77	2668.50	2253.86	0.71
BSA	2445.83	2494.82	2449.92	0.23
SGA	2250.77	2305.31	2251.02	0.13

Table 4
Probability (in %) that the SGA algorithm produces smaller minima than the ICA, BSA or FA, and its complements.

Algorithm	SGA is better (P_{better})	Alg. are equivalent (P_{eq})	Alternative is better (P_{worse})
FA*	0.20	0.00	**99.80***
ICA	32.83	63.16	4.01
BSA	100.00	0.00	0.00

* Best performance.

3.2. Eleven-bar truss, problem 1: size and topology optimization

Several authors have studied this problem, but employing two different sets of discrete design variables. For instance, Rajan [34] and Tang et al. [39] have adopted cross-sectional areas from a set of 32 discrete values, whereas Hajela et al. [19] and Deb & Gulati [7] have allowed the cross-sectional areas to vary within the range of 0.0–30.0 in^2 (0.0–193,55 cm^2) at increments of 1.0 in^2 (6.45 cm^2). The best known solution in the literature is 4912.15 lb 2228.11 kg), found by [6]; hence this is the problem version reproduced herein.

In this example, each algorithm run is limited to 30000 objective function evaluations (OFE). In order to provide sufficient data for the statistical analysis, the problem is run 1000 times with the FA, ICA, BSA and SGA algorithms. The parameters employed in these runs are presented in Table 2. The notation of parameters in this table is the same of the references of the algorithms, i.e. FA [11], ICA [1], BSA [37]and SGA [17]. This same notation is employed in the next examples.

Results for the thousand runs are summarized in Fig. 2 and in Tables 3 and 4. Fig. 2 shows the histograms of the thousand minima computed in thousand runs of each algorithm. Statistics usually computed, such as mean, standard deviation, best and worst minima are shown in Table 3. The proposed probability metrics are shown in Table 4, as well as their complements. Convergence plots are shown in Fig. 3 for the usual metrics and in Fig. 4 for the proposed probabilistic metrics.

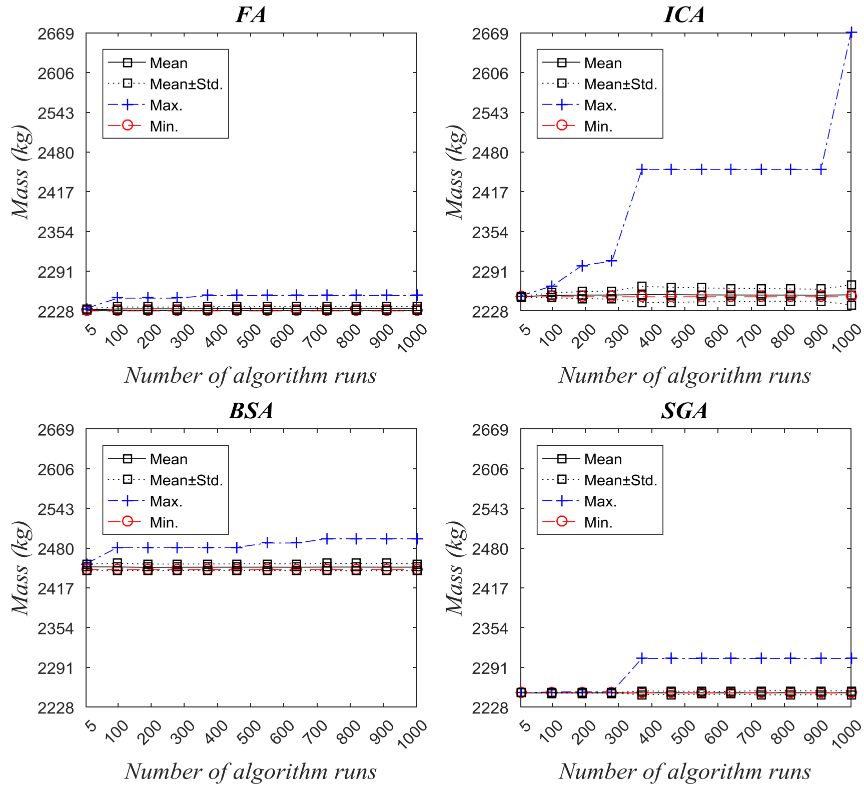

Fig. 3. Convergence histories for usual metrics in Problem 1.

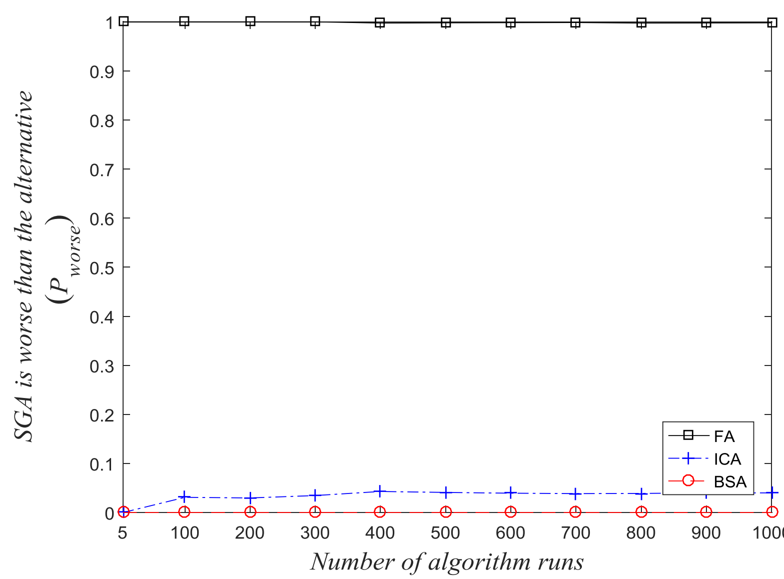

Fig. 4. Convergence histories for proposed probabilistic metric in Problem 1.

Table 5
Parameters of the FA, ICA, BSA and SGA for Problem 2.

Algorithms	Parameters
FA	$n_{pop} = 10$, $it_{max} = 5000$,
ICA	$N_{Col} = 214$, $N_{imp} = 3$, $\beta = 1.00$ and $\gamma = \frac{\pi}{4}$
BSA	$n_{pop} = 10$, $it_{max} = 5000$, $m_r = 1.0$, $\alpha \sim N(0, 3)$
SGA	$n_{pop} = 407$, $n_{mut} = 12$, $n_g = 37$, $\alpha^0 = 0.30$, $it^{max}_{global} = 0.50$

Table 3 shows that FA yielded the best (smallest) minima. However, this information is not sufficient to state that FA is superior to the other algorithms, for the following reasons: one also needs to look at the mean, the COV and the worst values obtained. Moreover, the values shown in Table 3, or the histogram shown in Fig. 2, are valid only for thousand runs of each algorithm. One also needs to look at convergence plots, which are not practical for the histograms of Fig. 2.

Looking again at Table 3, one readily sees that the mean and the worst values are also smaller for FA, as well as the COV (which in

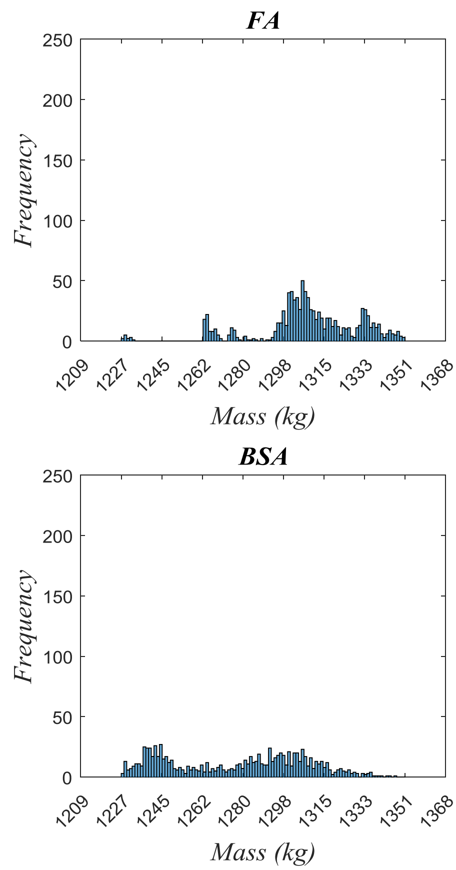

Fig. 5. Histograms of the best results over 1000 runs for Problem 2.

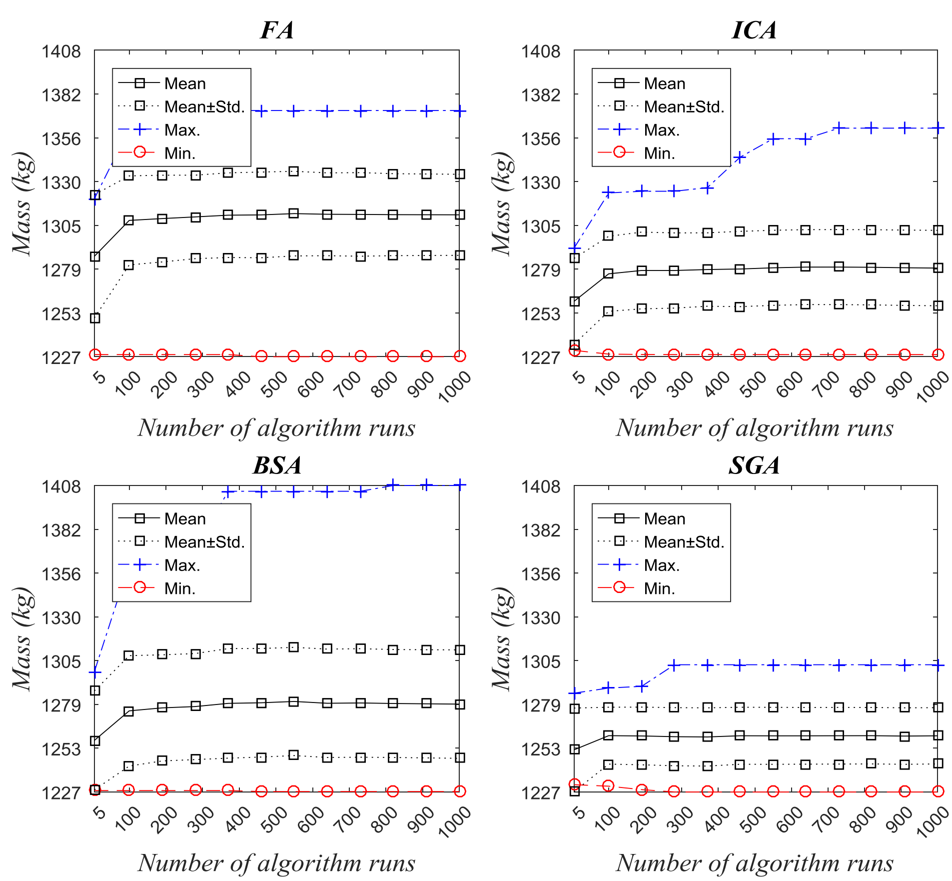

Fig. 6. Convergence histories for usual metrics in Problem 2.

Table 6
Statistics of objective function minima found over 1000 runs of each algorithm.

Algorithm	Best (kg)	Worst (kg)	mean (kg)	COV (%)
FA	1227.04	1372.37	1310.80	1.82
ICA	1228.31	1361.96	1279.38	1.74
BSA	1227.29	1407.94	1278.92	2.50
SGA	1227.04	1302.16	1260.22	1.32

this case is very similar to the COV of SGA). Fig. 3 indicates that indeed the results for FA (and for the other algorithms) very much converged at 1000 runs, hence confirming that FA is the best algorithm for this problem, with OFE limited to 30,000. Now looking at Table 4, it becomes immediately clear that the FA algorithm is much better than SGA: it is 99.8% more likely to produce the global minima in a single run! Looking at Fig. 4, it becomes immediately clear that the FA algorithm is better than SGA, for this problem, from the fifth run.

3.3. Eleven-bar truss, problem 2: size, shape and topology optimization

In this example, shape is also optimized by allowing the vertical coordinates of the three superior nodes to move between 180 in 4.572 m) and 1000 in 25.4 m), considering the origin in the intersection of members 1, 2, and 3. Because the nodal coordinates are continuous and the cross-sectional areas are taken from a set of 32 discrete variables [34,11]; [17], the problem is a mixed variable optimization problem in that it addresses integer and continuous design variables simultaneously.

In this example, each algorithm run is limited to 50 thousand OFE. As in the previous example, this problem was run 1000 times with the FA, ICA, BSA and SGA. The parameters of each algorithm employed in this analysis are given in Table 5.

The histograms of minima obtained for a thousand runs of each algorithm are compared in Fig. 5. It is observed that all algorithms identified the minima eventually, but the dispersion of SGA is smaller. This hints that SGA is the best algorithm for this problem, but this is valid only for a thousand runs and 50 thousand OFEs. Results in Table 6 confirm that SGA is better for this problem, in this setting, since the mean, the worst and the COV are smaller than for the other algorithms. Fig. 7 confirms that SGA is better for any number of algorithm runs.

Table 7
Probability (in %) that the SGA algorithm produces smaller minima than the ICA, BSA or FA, and its complements.

Algorithm	SGA is better (P_{better})	Alg. are equivalent (P_{eq})	Alternative is better (P_{worse})
FA	93.90	0.00	6.10
ICA	78.61	0.00	21.39
BSA	69.03	0.00	30.97

Now, how much better is SGA than the other algorithms? Marginally or significantly? Results in Table 6, or in Fig. 5 or 6, do not say it. In Table 7, it is clearly stated that SGA presents a probability of 93.90% of producing smaller minima than FA, in a single run, while for ICA and BSA the probabilities are 84.17% and 79.35%. Results in Table 6 and Fig. 5 or 6 also do not clearly show which algorithm is better, excluding SGA. Table 7 and Fig. 7 clearly show that ICA and BSA are better than FA, for 50 thousand OFEs.

What if a different number of objective function evaluations (OFE) were allowed for each algorithm? It would be confusing to compare numbers like Table 6, or convergence plots like Fig. 6. However, results are clear in Fig. 8, where it is immediately seen that SGA is also better than FA also for 10 or 30 thousand OFEs. Similar results were also computed w.r.t. ICA and BSA, but are not shown herein.

3.4. 25-bar 3D truss: size, shape and topology optimization

This 3D truss is also often used as a benchmark problem [21,16]. The ground structure is shown in Fig. 9 and the details of the loading and member groupings are given in Tables 8 and 9. The allowable strength is 275.79 MPa in tension and compression, the material properties (modulus of elasticity and weight density) are the same as in the previous problems. The maximum allowable deflection is 0.889 cm in any direction for each node. The cross-sectional areas are chosen from a set $D = (0.254, 0.508, 0.762, 1.016, 1.270, 1.524, 1.778, 2.032, 2.286, 2.540, 2.794, 3.048, 3.302, 3.556, 3.810, 4.064, 4.318, 4.572, 4.826, 5.080, 5.334, 5.588, 5.842, 6.096, 6.350, 6.604, 7.112, 7.620, 8.128, 8.636)$ cm.

The *x*-, *y*- and *z*-coordinates of nodes 3, 4, 5 and 6 and the *x*- and *y*-coordinates of nodes 7, 8, 9 and 10 are taken as design variables, while nodes 1 and 2 remains unchanged. Because double symmetry is required in both the *x*-*z* and *y*-*z* planes, the problem includes

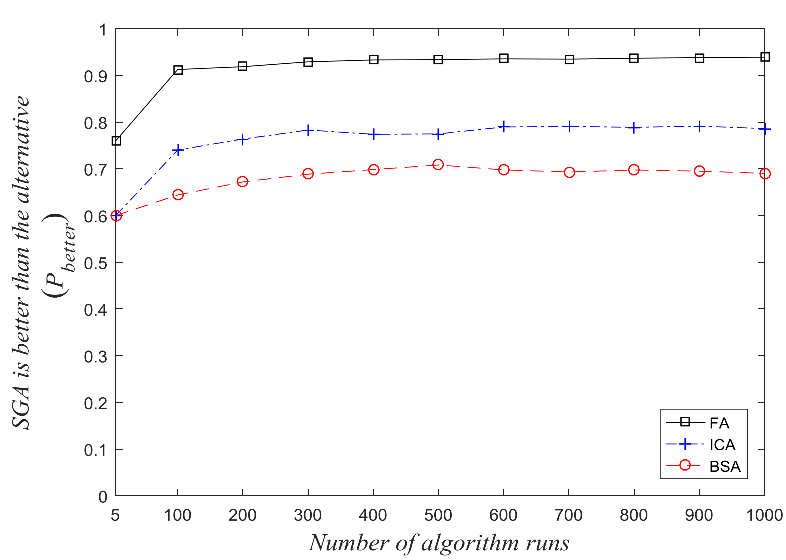

Fig. 7. Convergence histories for probabilistic metric in Problem 2, comparison of algorithms.

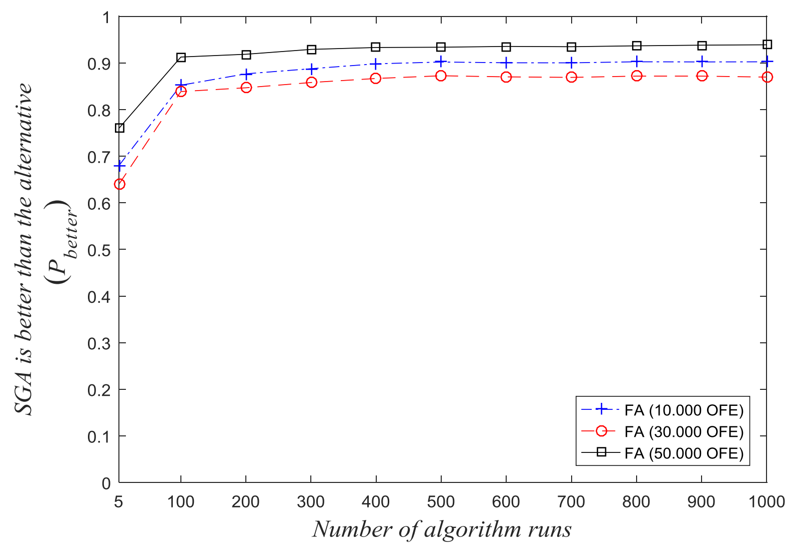

Fig. 8. Convergence histories for probabilistic metric in Problem 2, different numbers of objective function evaluations (OFE).

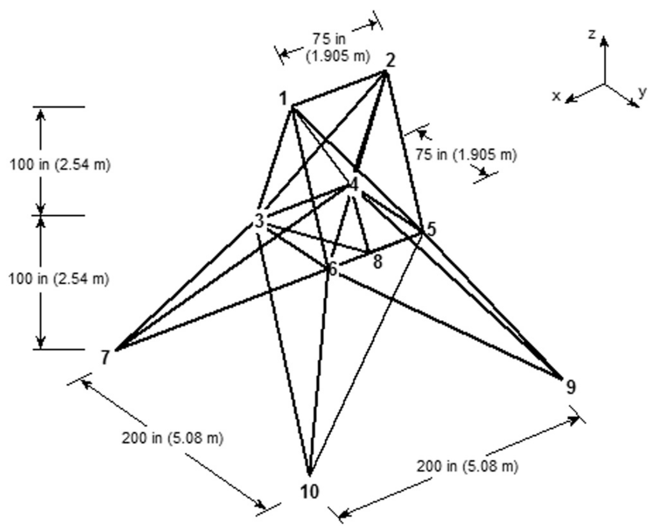

Fig. 9. Twenty-five 3D truss benchmark example, Problem 3.

Table 8
Loading of twenty-five bar 3D truss.

Node	P_x (kN)	P_y (kN)	P_z (kN)
1	4.4482	−44.4822	−44.4822
2	0	−44.4822	−44.4822
3	2.2241	0	0
6	2.6689	0	0

eight size and five configuration variables. The side constraints for the configuration variables are 50.8 cm $\le x_4 \le$ 152.4 cm, 101.6 cm $\le x_8 \le$ 203.2 cm, 101.6 cm $\le y_4 \le$ 203.2 cm, 254 cm $\le y_8 \le$ 355.6 cm and 228.6 cm $\le z_4 \le$ 330.2 cm. Because the nodal coordinates are continuous and the cross-sectional areas are taken from a set of 30 discrete variables, this problem is also a mixed variable optimisation problem in that it deals simultaneously with integer and continuous design variables. In addition to the eight discrete size and five continuous configuration variables, all eight member groups are considered as topology variables.

The algorithms FA, ICA, BSA and SGA were run 1000 times for this problem, with each run limited to 6 thousand OFE. The histograms of these runs are illustrated in Fig. 10, while the classical statistical analysis and the approach proposed in this paper are presented in Tables 11 and 12, respectively.

Looking at the histograms of Fig. 10, it is easy to see that BSA and SGA are better than FA and ICA, for one thousand runs and 6 thousand OFE. However, it is not possible to say which algorithm in better between BSA and SGA. Looking at the conventional statistics in Table 10 is a little puzzling, because SGA resulted in the best minima, with the smallest COV and worst value, but BSA produced an almost-as-small minimum, with smaller mean. The probabilities in Table 12 give a slight advantage to BSA, in this case, as it is almost 60% more likely to produce the global minima in a single run, in comparison to SGA.

Figs. 11 and 12 show the convergence histories for the usual metrics and for the probabilistic metric proposed herein. Again, it is very difficult to compare the performance of BSA and SGA in Fig. 11. However, Fig. 12 confirms that BSA is better than SGA from

Table 9
Node co-ordinates and member grouping of twenty-five bar 3D truss.

Node	x (cm)	y (cm)	z (cm)	Group	Member (end nodes)
1	−95.25	0	508	A1	1(1,2)
2	95.25	0	508	A2	2(1,4), 3(2,3), 4(1,5), 5(2,6)
3	−95.25	95.25	254	A3	6(2,5), 7(2,4), 8(1,3), 9(1,6)
4	95.25	95.25	254	A4	10(3,6), 11(4,5)
5	95.25	−95.25	254	A5	12(3,4), 13(5,6)
6	−95.25	−95.25	254	A6	14(3,10), 15(6,7), 16(4,9), 17(5,8)
7	−254	254	0	A7	18(3,8), 19(4,7), 20(6,9), 21(5,10)
8	254	254	0	A8	22(3,7), 23(4,8), 24(5,9), 25(6,10)
9	254	−254	0		
10	−254	−254	0		

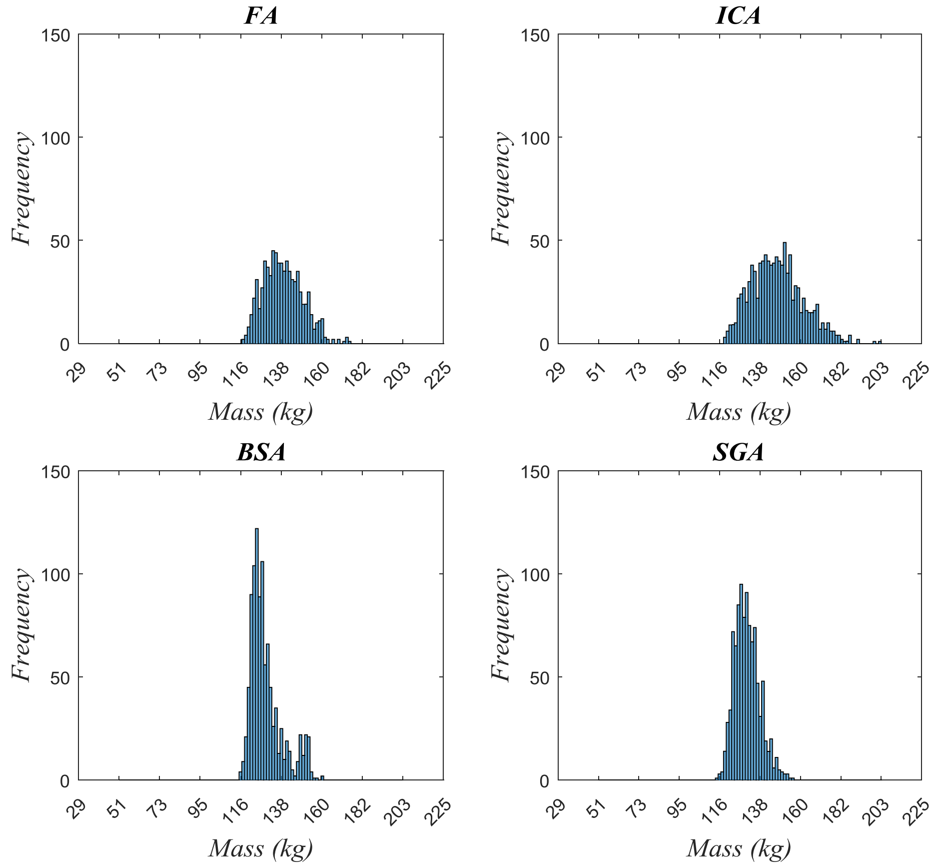

Fig. 10. Histograms of objective function minima computed in 1000 runs of Problem 3.

Table 10
Parameters of the FA, ICA, BSA and SGA for Problem 3.

Algorithms	Parameters
FA	$n_{pop} = 10$, $it_{max} = 600$
ICA	$N_{Col} = 214$, $N_{imp} = 3$, $\beta = 1.00$ and $\gamma = \frac{\pi}{4}$
BSA	$n_{pop} = 10$, $it_{max} = 600$, $m_r = 1.0$, $\alpha \sim G_a(1,1)$[4]
SGA	$n_{pop} = 50$, $n_{mut} = 3$, $n_g = 8$, $\alpha^0 = 0.25$, $it_{global}^{max} = 0.20$

[4] $G_a(1,1)$: scalar random value chosen from a gamma distribution with unit scale and shape.

Table 11
Statistics of objective function minima found over 1000 runs of each algorithm.

Algorithm	Best (kg)	Worst (kg)	Mean (kg)	COV (%)
FA	117.55	174.70	139.05	7.48
ICA	118.86	203.24	147.29	9.50
BSA	115.72	160.20	129.89	6.70
SGA	114.45	155.78	131.11	5.11

Table 12
Probability (in %) that the SGA algorithm produces smaller minima than the ICA, BSA or FA, and its complements.

Algorithm	SGA is better (P_{better})	Alternative is better (P_{worse})
FA	79.46	20.54
ICA	85.23	14.77
BSA[*]	40.55	**59.45**[*]

[*] Best performance.

the onset, i.e., also for a smaller number of algorithm runs. It is also easy to see that FA is better than ICA for this problem, especially for small number of runs.

The above results are strictly valid only for the six thousand objective function evaluations (OFE) specified for each algorithm run. Fig. 13 shows that the FA algorithm is actually better than SGA if algorithm runs are limited to one thousand OFEs, and marginally better for three thousand OFEs, for any number of algorithm runs.

It becomes clear that a proper comparison between algorithms needs to consider the number of objective function evaluations considered, and the number of runs for each algorithm. Such comparisons are virtually impossible to perform, using usual metrics such as those shown in Fig.11, or histograms such as shown in Fig.10.

4. Concluding remarks

In this paper, a new method to compare the performance of metaheuristic algorithms was proposed. The method is based on the idea of statistical interference, and yields the probability that a given (proposed?) algorithm produces a smaller (global?) minimum than an alternative algorithm. The so-called $\boldsymbol{P_{better}}$ metric was employed in the solution of three benchmark examples from the literature, which were solved using four optimization algorithms. It was shown that the proposed metric allows one to make clear statements about the advantages of one algorithm over the other. This includes a quantifiable measure of how much a given algorithm is better than another.

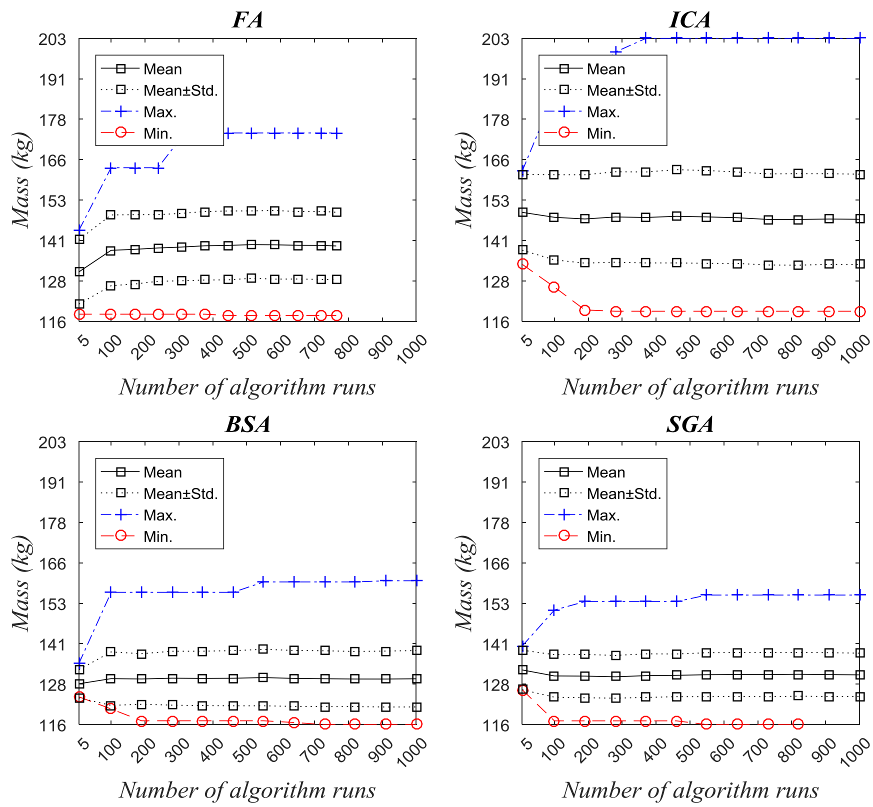

Fig. 11. Convergence histories for usual metrics in Problem 3.

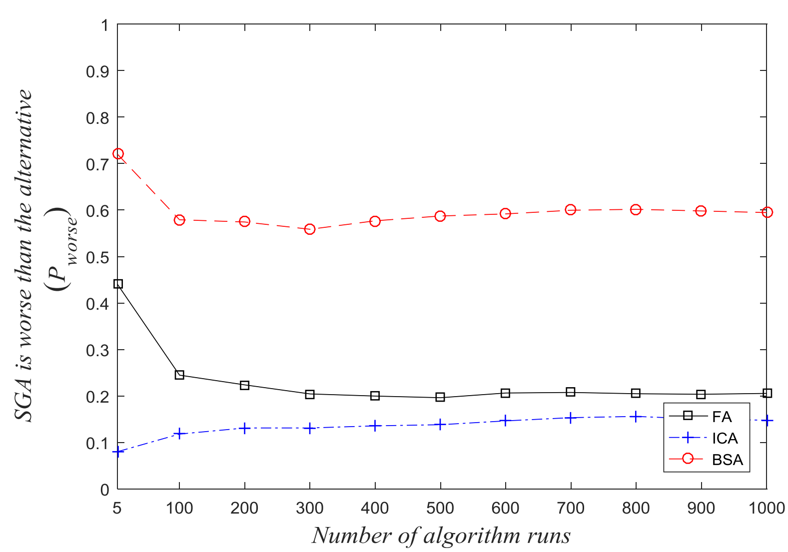

Fig. 12. Convergence histories for proposed probabilistic metric in Problem 3, comparison of algorithms.

The present study has also shown that a proper comparison between metaheuristic algorithms requires consideration of the number of objective function evaluations for each run, and the number of runs of each algorithm. Such comparisons are virtually impossible to perform with the usual metrics such as mean, coefficient of variation, minimum and maximum values obtained in several run (as shown in Figs. 3, 6, 11), or using histograms (as shown in Figs. 2, 5, 10). The single probabilistic $\boldsymbol{P_{better}}$ metric proposed herein can be easily computed for different numbers of objective function evaluations and runs, yielding a more comprehensive comparison between algorithms.

Out of four algorithms tested, one was found to be the best for each of three problems tested! Hence, we confirm herein that algorithm performance is very problem dependent: this only helps to make the case for the necessity of proper tools for evaluating the performance of optimization algorithms.

The $\boldsymbol{P_{better}}$ metric proposed herein is certainty only one of many acceptable metrics. Herein, it was only compared to the usual proce-

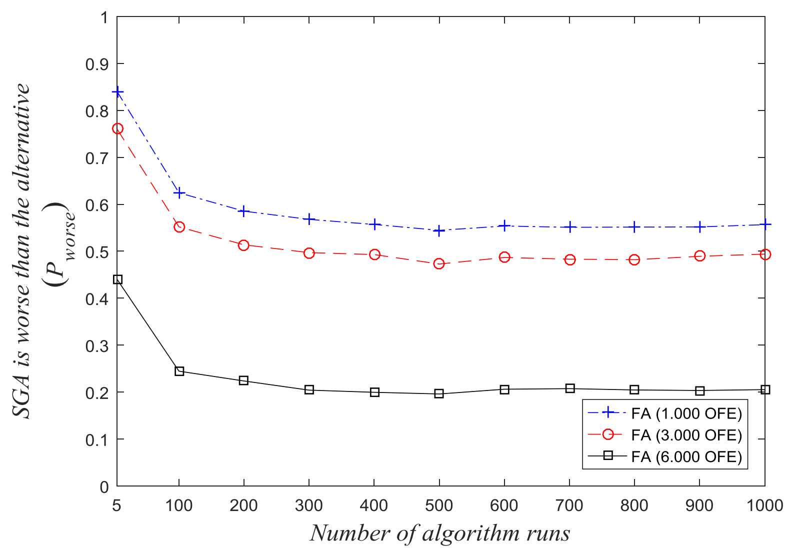

Fig. 13. Convergence histories for proposed probabilistic metric in Problem 3, comparison of objective function evaluations (OFE).

dure of computing best and worst values, and mean and coefficient of variation of the minima generated in different runs. As usual for non-convex problems solved using heuristic algorithms, there is no warranty that the resulting minima are indeed global minima.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.strusafe.2017.10.006.

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